

Exploring parametric uncertainty using initialized predictions with the Community Earth System Model

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catalyst

Sources of model error

$$\psi^t = \psi^{t-1} + \hat{f}(\psi^{t-1}, \hat{\mathbf{x}}; \mathbf{p})$$

Prediction

Prior value

Model physics

$$\varepsilon_{struct} = f(\mathbf{x}) - \hat{f}(\hat{\mathbf{x}})$$

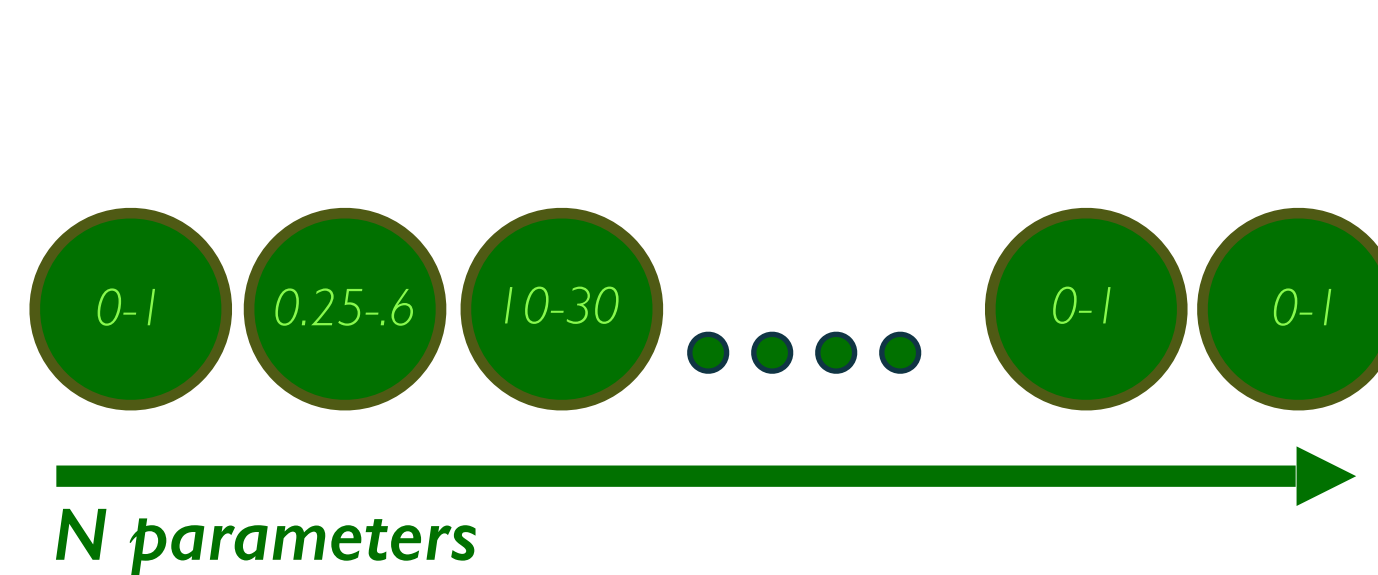
Real-world physics

Model physics

Uncertainty Source	Question	Representation	Example
Initial Condition	"Where are we starting?"	State estimate	Analysis / Data Assimilation
Boundary Condition	"What's pushing from outside?"	Forcing terms	Sea surface temperatures or solar radiation.
Parametric	"How strong is this specific effect?"	Knobs (adjustable constants)	The coefficient for cloud droplet autoconversion.
Structural	"Are we using the right physics?"	Equations	Empirical linear fit to lab measurements

Parametric uncertainty & PPEs

the uncertainties in *input parameters* which are addressed by varying input values within a defined range to assess their impact on outcomes.



microphysics (DCS)

$$\lambda_{ice} \leq \frac{1}{2} DCS$$

$$D_{ratio} = \lambda_{ice} \times DCS$$

deep convection (CAPELMT)

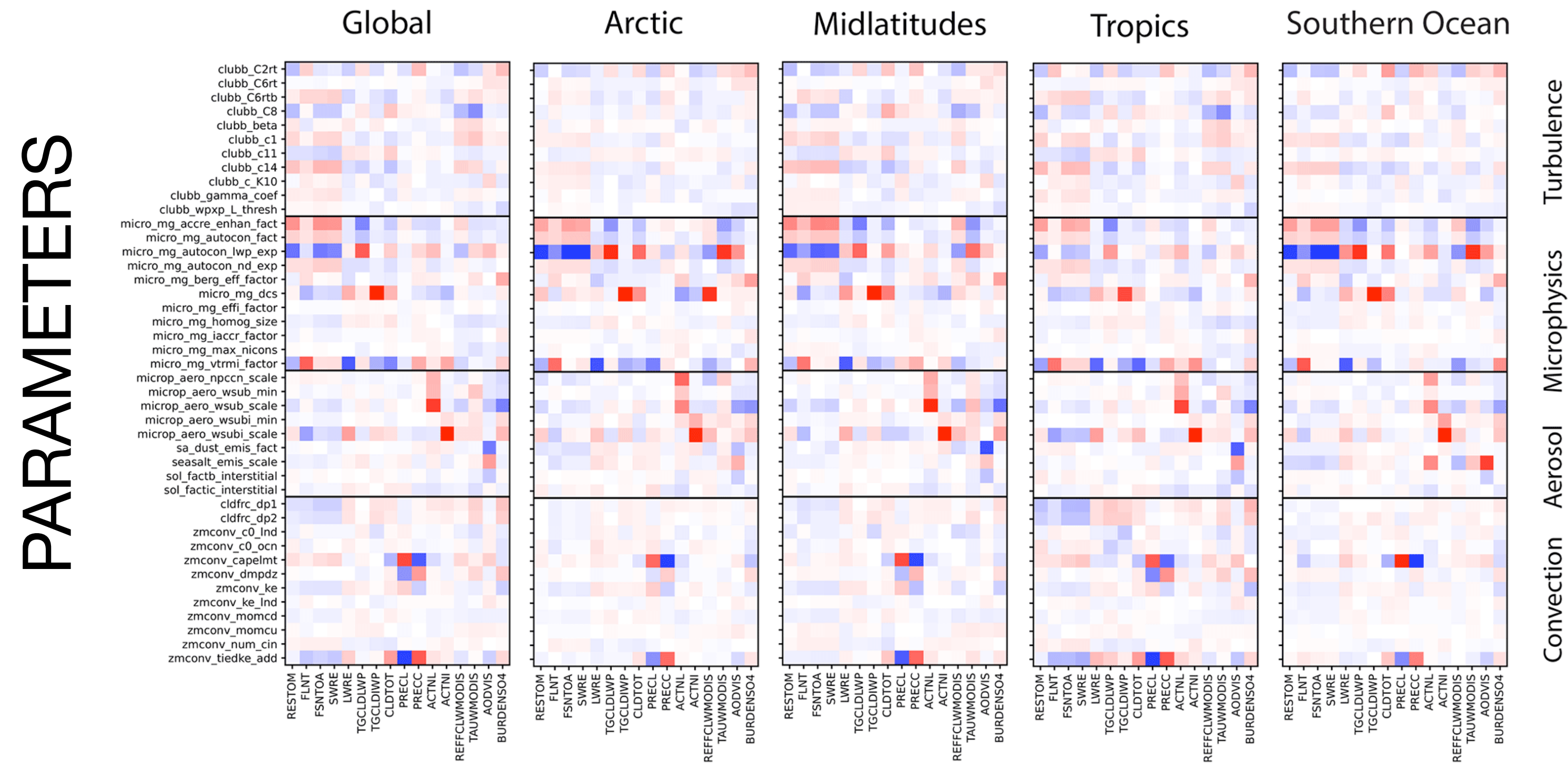
$$\mathfrak{D} = \begin{cases} 1, & \text{if } CAPE > CAPE_{LMT} \\ 0, & \text{otherwise} \end{cases}$$

$$M_b = \frac{(CAPE - CAPE_{LMT})}{\tau \frac{da}{dt}}$$

Parametric uncertainty & PPEs

CAM6 Perturbed Parameter Ensemble

- Randomly sample 45 parameters within specified ranges
- ~260 simulations, 3-years each



iPPE

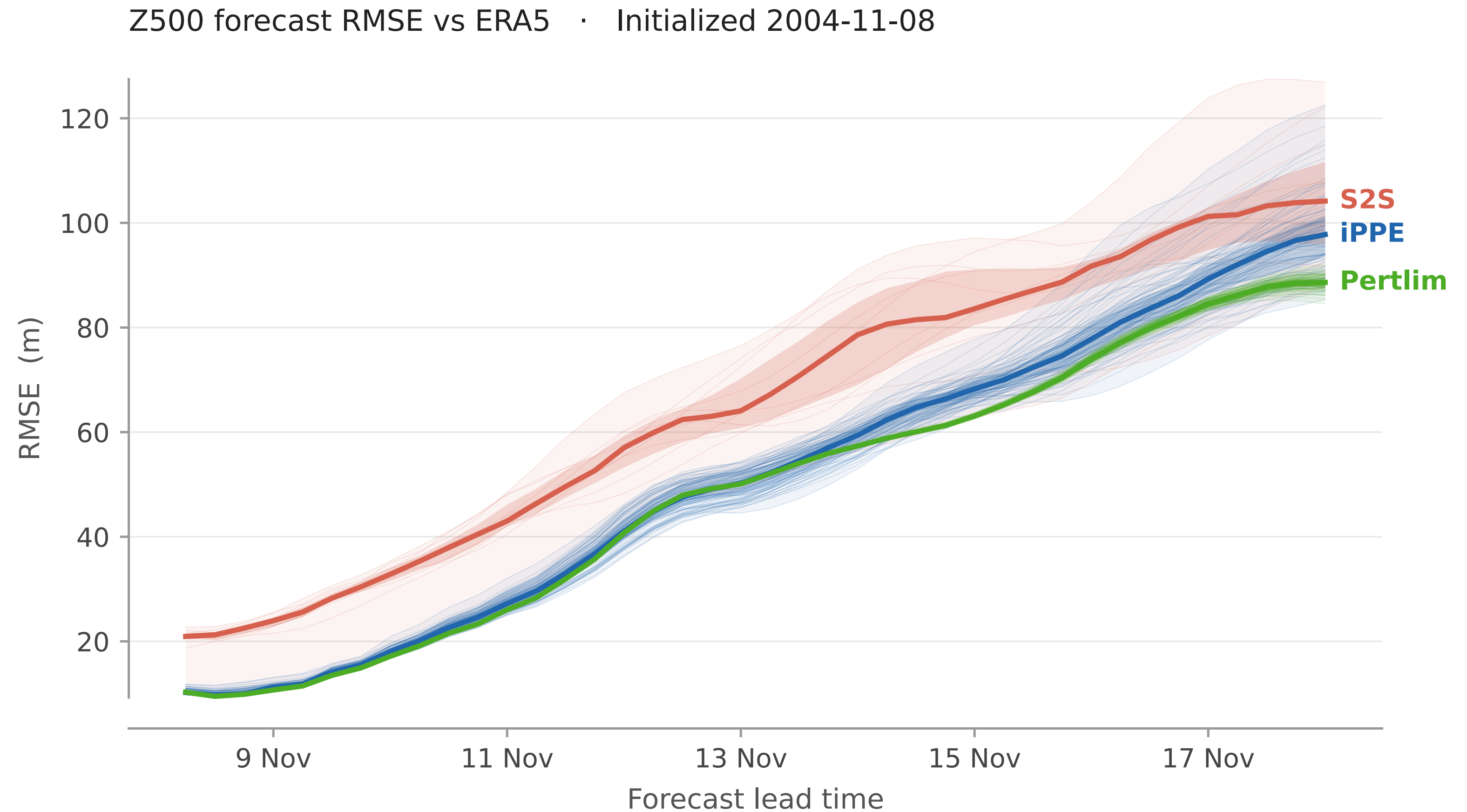
An initialized PPE

- ▶ Questions
 - Can individual forecasts be optimized by parameter “tuning”?
 - Do optimal parameter values vary with the atmospheric state?
 - Do optimal parameter values from short weather forecasts differ from climate settings?
- ▶ Setup
 - Initialize from same atmosphere/land IC as S2S,
 - But *not* coupled to ocean (observed SST/ice)
 - CAM6 (1°, 32-levels)
 - 10-day forecast, 100 members, ~35 parameters

iPPE preliminary results

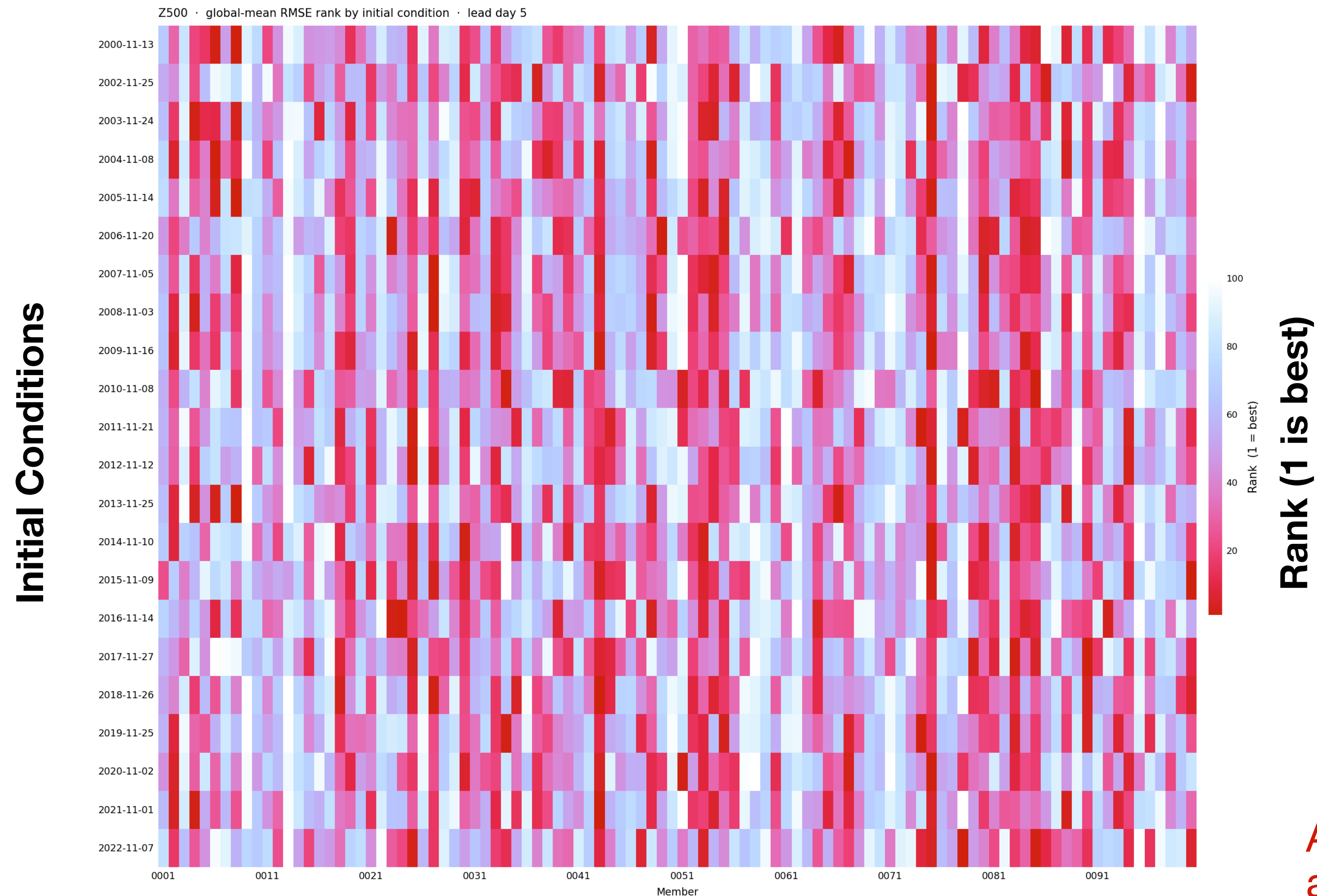
Global Z₅₀₀ RMSE

- ▶ Common skill indicator (e.g., [Buizza & Leutbecher 2015](#))
- ▶ Example from one initial condition
- ▶ Comparison:
 - **S2S**
large initial perturbation, coupled, n=11
 - **Pertlim**
random noise perturbation; default parameters; same SST/ice as iPPE (n=100)



iPPE preliminary results

Do some parameter sets outperform others?



- ▶ 22 initial conditions, all from November (2000-2022)
- ▶ 2,200 simulations (22 IC × 100 parameter sets)

A red column would indicate a parameter set with lowest error for all cases.

iPPE preliminary results

Kendall's W is a non-parametric statistic for rank correlation.

- Rank the Z_{500} RMSE across the n ensemble members for each of the m initial conditions, define the total rank:

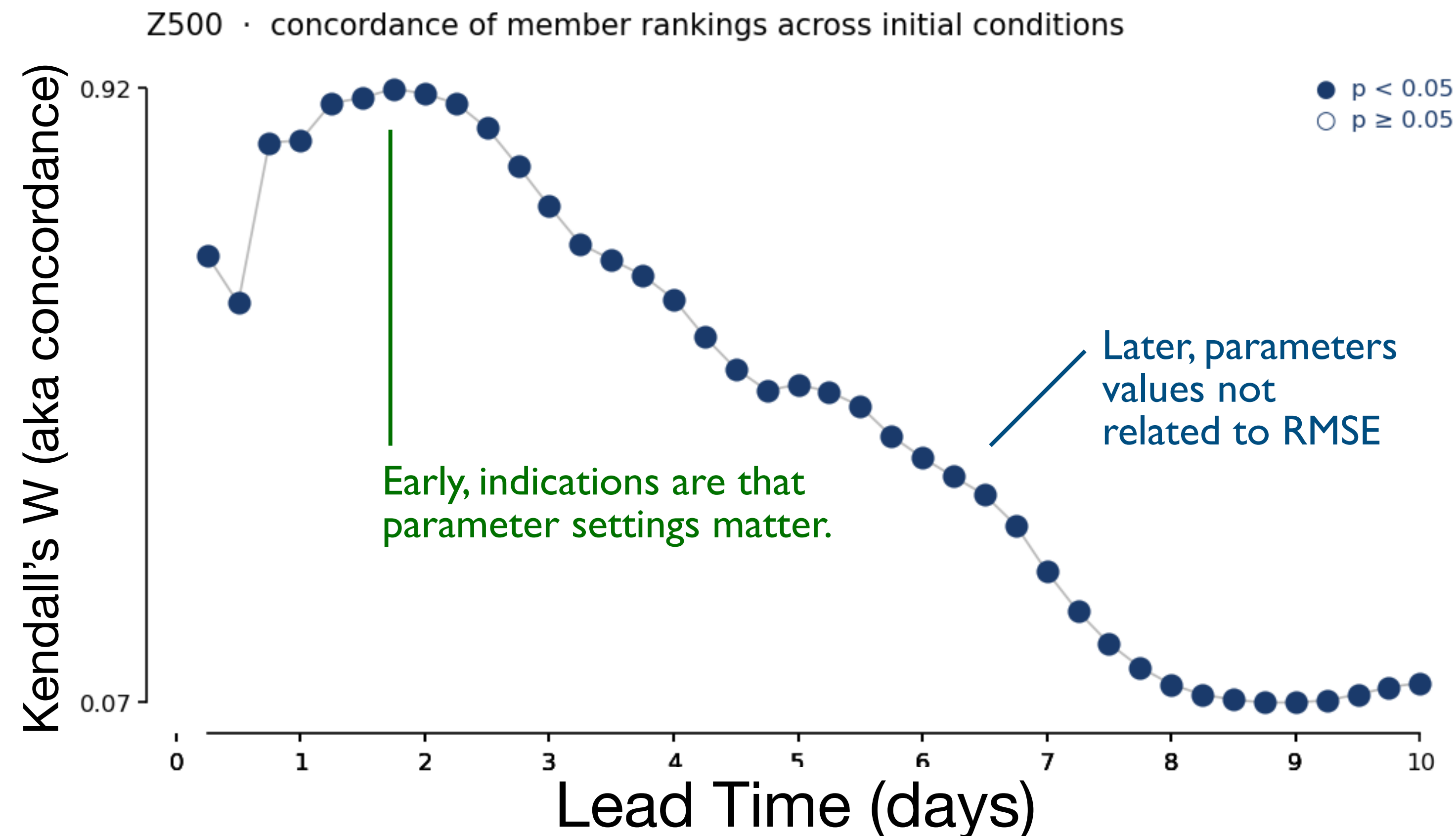
$$R_i = \sum_{j=1}^m r_{i,j}$$

- Sum of squared deviations from the mean rank:

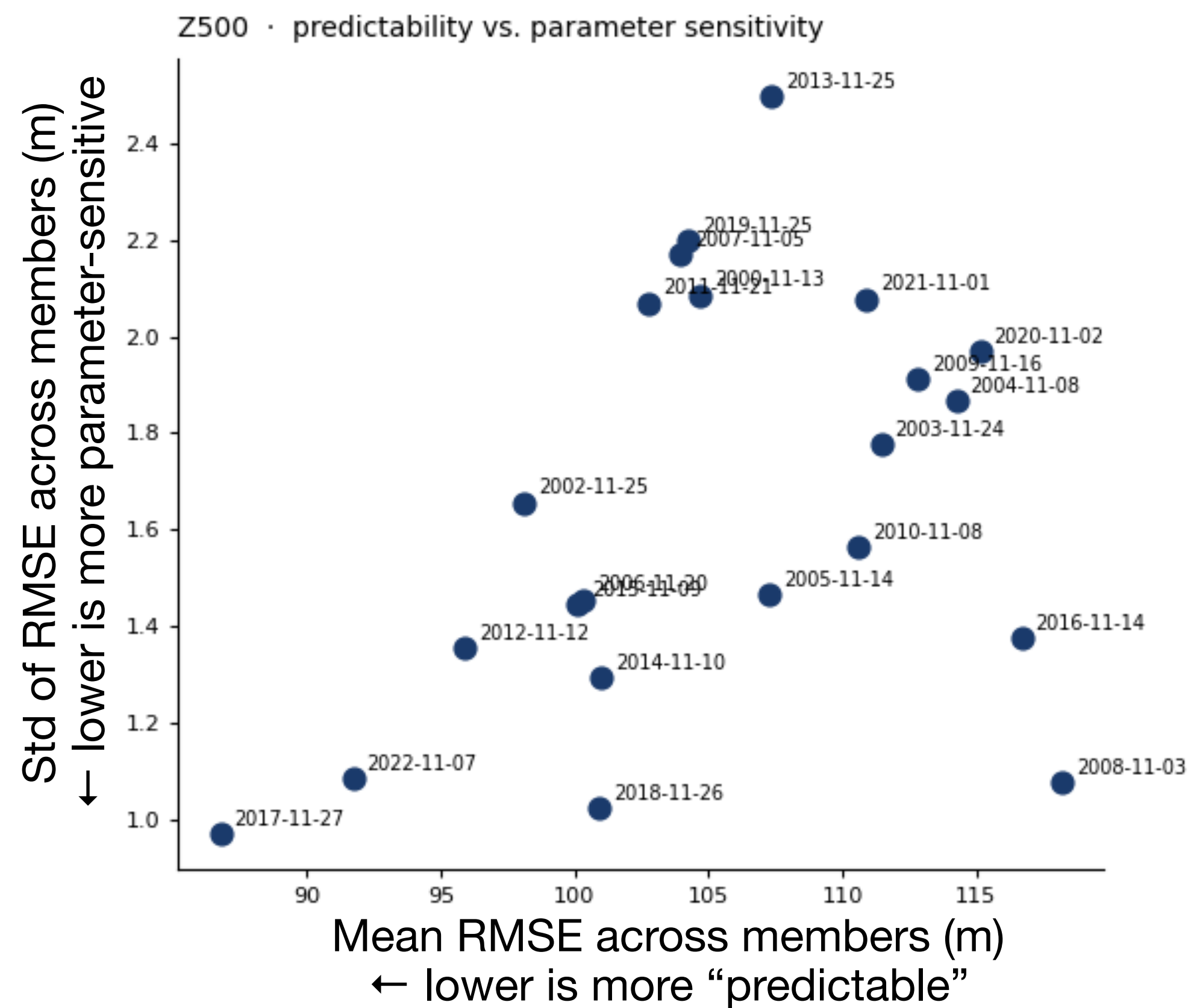
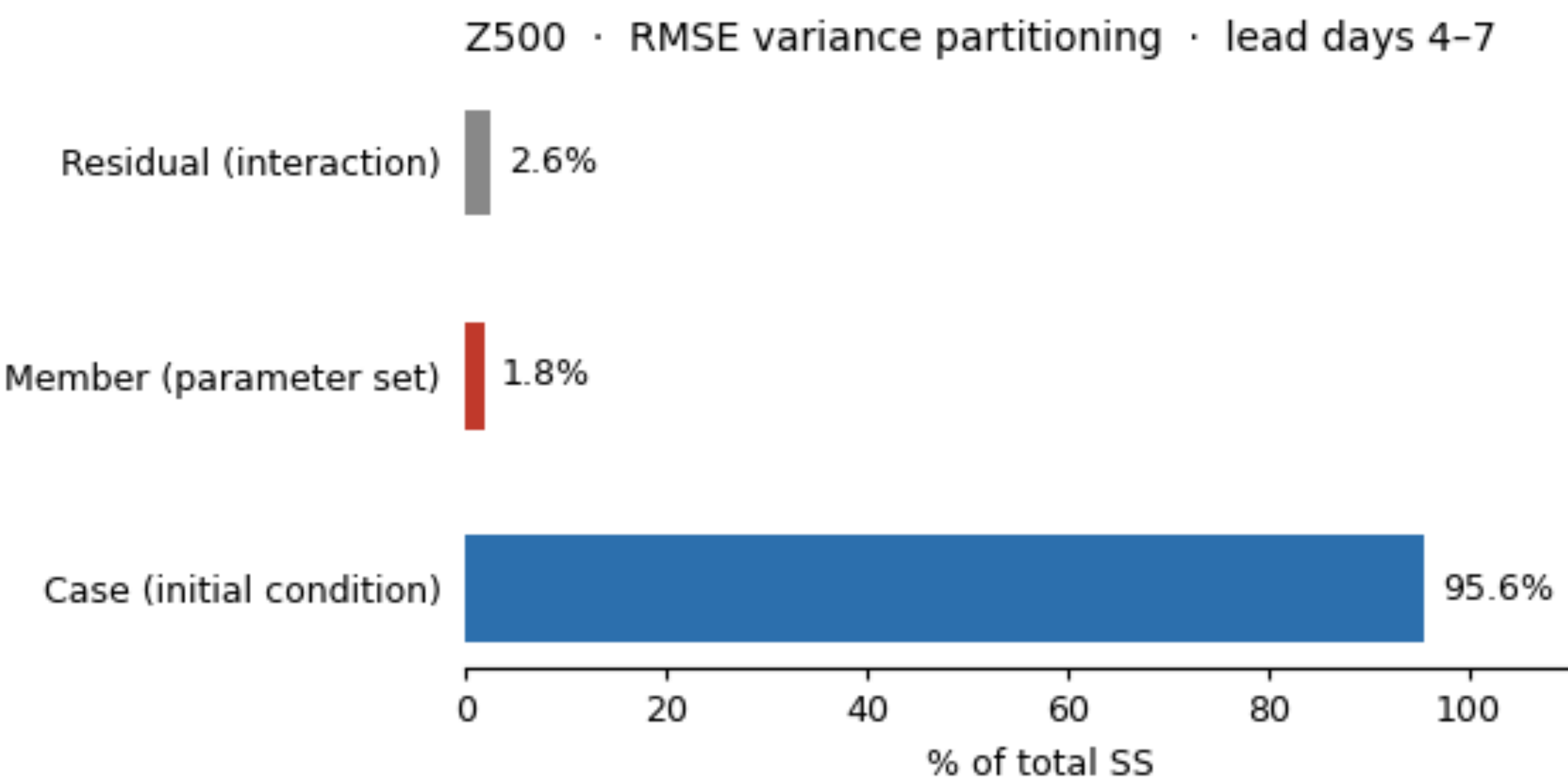
$$S = \sum_{i=1}^n (R_i - \bar{R})^2$$

- W then is defined:

$$W = \frac{12S}{m^2 (n^3 - n)}$$

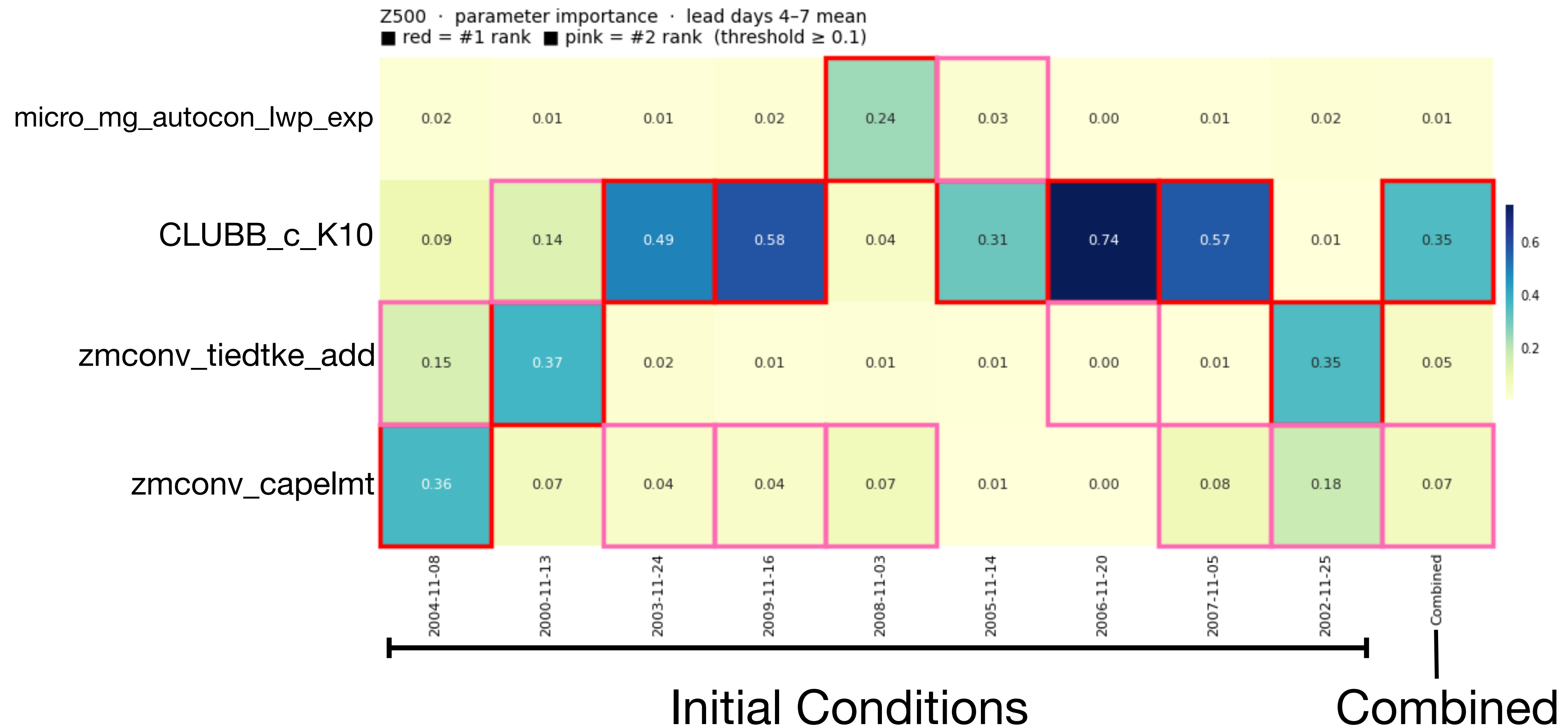


Exploring state dependence



What parameters impact skill?

Random forest regressor, 1000 estimators



Conclusion

► Preliminary Findings

- Parameter settings impact forecast trajectory ✓
- Different parameter sets are “best” for different initial conditions ✓
 - ... but sometimes only after ~2 days ✓
- Within a few days, most variance related to IC ✗
 - ... but some IC ensembles show significant parameter sensitivity ✓

► Ongoing

- Extend analysis to T_{2m} , precipitation, and/or other variables/regions of interest
- Refine parameter calibration method to answer if optimal parameters vary by initial state.